Proposal Defense

Socratic AI, Agentic AI for Fostering Critical Thinking Education

A multi-agent tutor that asks instead of answers, and measures how well the learner reasons

Presented by Muhammad Eusoff Amir bin Hazri (22010491)

Supervised by Dr Ho Tatt Wei

Internal examiner Dr Syaimaa' Solehah M Radzi

Department of Electrical and Electronics Engineering

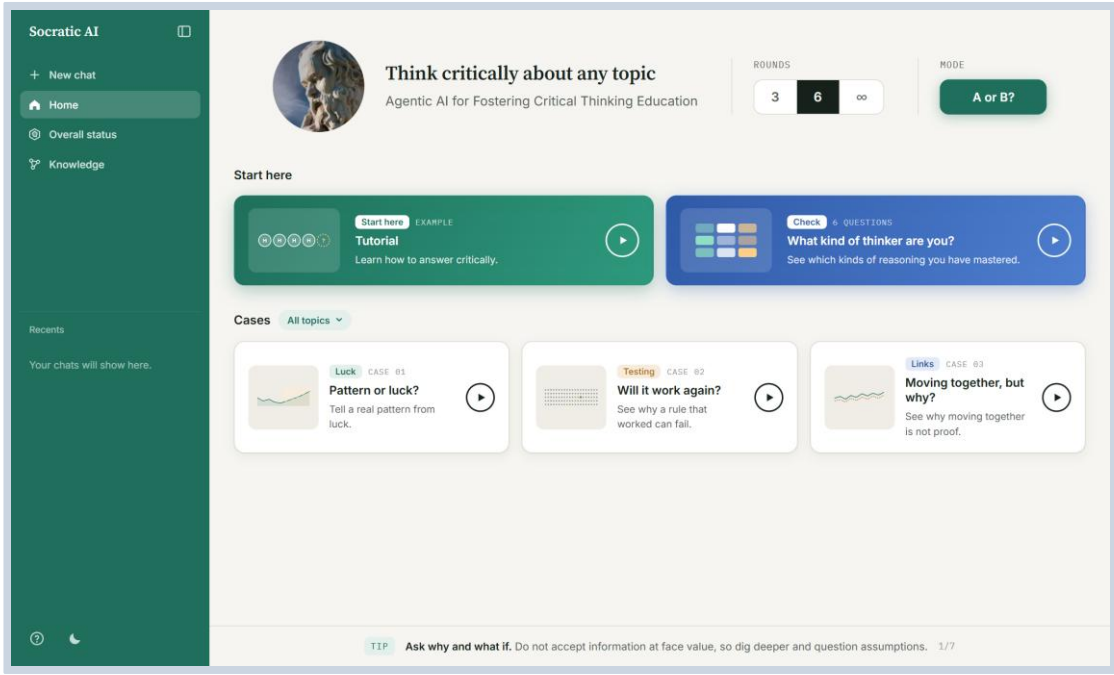
Course EFB4042 Final Year Project I

Project code 26S005

Date 28 October 2026

OUTLINE OF PRESENTATION

1. Introduction
2. Problem Statement, Objectives, Scope of Study
3. Literature Review
4. Proposed Methodology
5. Gantt Chart & Milestones
6. Preliminary Results
7. Conclusion



The working prototype, live online

INTRODUCTION

Students can now ask AI for an answer and copy it, so some skip the thinking.

62% of students

use generative AI at least weekly [12]

319 workers studied

says more trust in AI meant less critical thinking [10]

4 costs of "outsourced reasoning"

- fewer ideas,
- less effort,
- weaker memory,
- less reflection [11]

The idea of this project

Turn the AI around. The tutor asks questions in a set order, so the student does the thinking, and then it measures that thinking.

PROBLEM STATEMENT

AI now does much of the analysis and coding, so engineering students need strong critical thinking [10]. Socratic AI tutors avoid giving answers, but they have three gaps.

1

No ladder

The questions only lead to an answer, never to whether it is the right question.

2

No measurement

They do not measure the learner's critical thinking against any framework.

3

No adaptation

They do not adapt to what each learner is missing. One tester wrote "I've explained this before."

For example, a student asks a chatbot why a circuit keeps overheating, copies the confident fix, and never asks what caused it.

So engineering students need an agentic AI tool that builds their critical thinking.

OBJECTIVES

1. Undergraduates who are new to a body of knowledge reach higher levels of Paul's six levels of Socratic questioning, without the tutor giving them the answer.

2. The tutor's marks of each learner's critical thinking, on Paul and Elder's nine standards and on the type of reasoning and knowledge used, agree with human markers.

3. Each learner gets questions and hints that match their gaps, and can see which skills they have mastered and which still need work.

Each objective can be judged as met or not met at the end of the project.

SCOPE OF STUDY

1

Learners

Beginner undergraduates at UTP.
Short English text chats.

2

Knowledge

Any body of knowledge as text.
A few topics from different fields.
Quant checked by hand.

3

Critical thinking

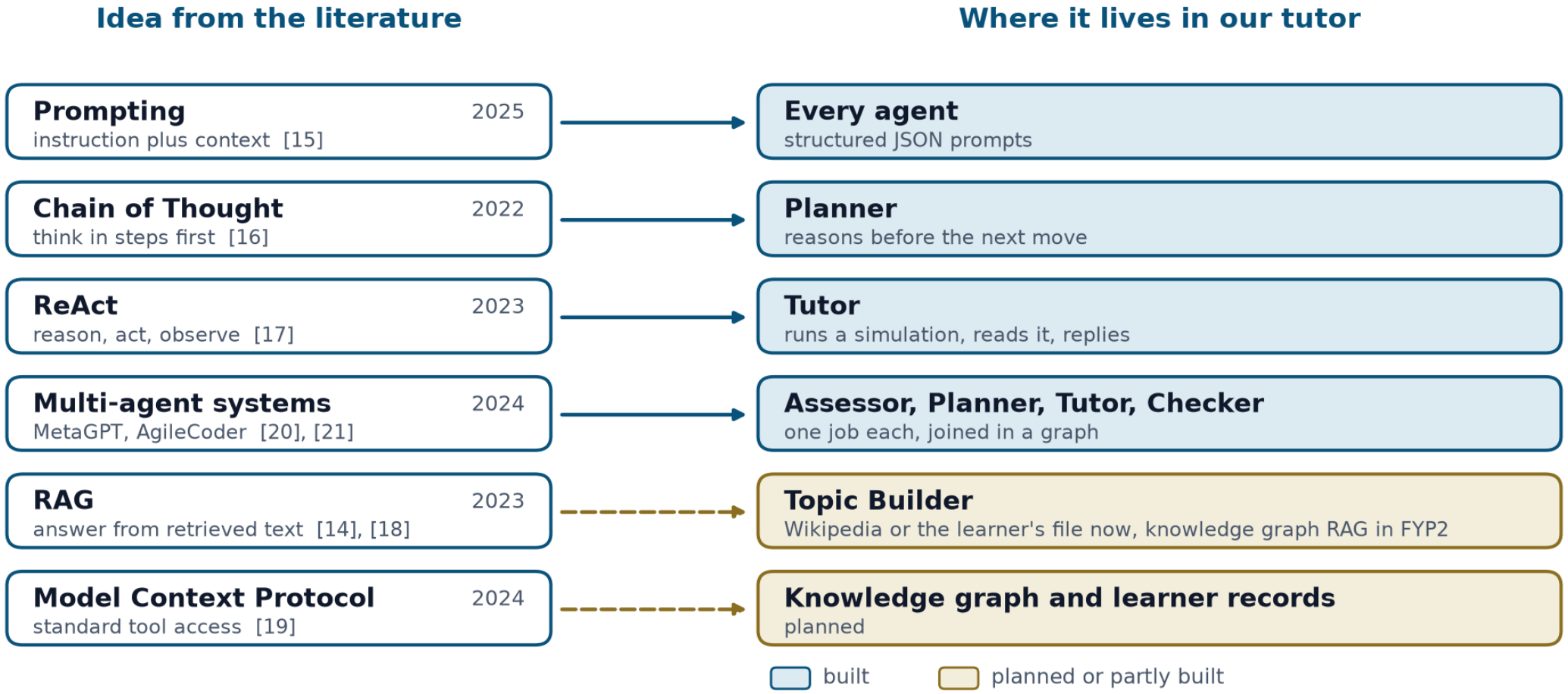
As defined by Paul and Elder.
Measured only inside the sessions.

Outside the scope are long-term change in how learners think and a full class study.

These belong to the later phases of the parent grant.

LITERATURE REVIEW

Agentic AI Building Blocks



Based on Dr Ho's course slides [15], [22] and the DeepLearning.AI courses [23].

LITERATURE REVIEW

Critical Thinking, What We Ask and Mark

Learner says "The circuit overheats because the resistor is too small."

6 levels, what the Tutor asks next

6	**Question the question**	Is overheating the real problem?
5	**Implications**	If you swap it, what else changes?
4	**Viewpoints**	What else could make it overheat?
3	**Evidence**	How could you check it is the resistor?
2	**Assumptions**	What are you assuming about it?
1	**Clarify**	What do you mean by too small?

9 standards, what the Assessor marks

Clarity Can you give an example?	**Accuracy** Is that true?	**Precision** Can you be more exact?
Relevance How does it relate?	**Depth** What makes it hard?	**Breadth** Is there another view?
Logic Does it follow?	**Significance** Is it the key point?	**Fairness** Is it fair to other views?

Each standard is marked 1 (met) or 0 (not met), with evidence from the answer

LITERATURE REVIEW

Types of Reasoning and Knowledge

3 types of reasoning

Inductive *many cases, then a general idea*

Every board from this batch overheats,
so the batch may share a fault.

Deductive *a rule, then a sure conclusion*

The resistor is rated 0.25 W but takes 0.5 W,
so it will overheat.

Abductive *what was seen, then the best explanation*

It only overheats after an hour, so heat
building up slowly is the likely cause.

8 types of knowledge, and the question each answers

Factual
What is true?

Conceptual
What does it mean?

Procedural
How do I do it?

Causal
Why does it happen?

Relational
How are things connected?

Strategic
What should I do here?

Metacognitive
How do I know, how sure am I?

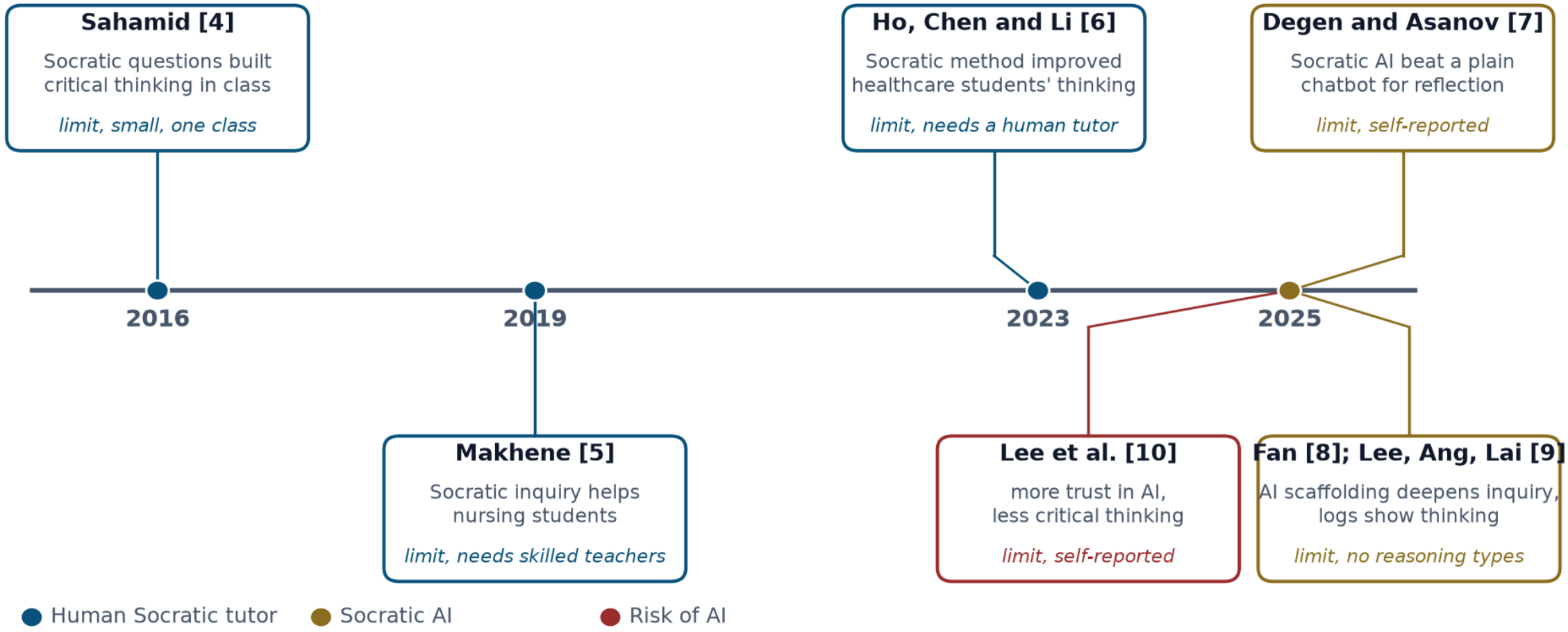
Epistemic
What is it fair to believe?

White boxes are knowing. Gold boxes lead to a decision or a judgement.

Knowledge types extend revised Bloom [2], and SOLO [3] will grade how many types an answer joins.

LITERATURE REVIEW

Socratic Questioning and Socratic AI



LITERATURE REVIEW

Existing Socratic AI Tutors

	Refuses answers	Question ladder	Framework marks	Goal, progress, end	Reasoning and knowledge types
Gemini Gem Socratic Tutor (Test #1)	●	○	○	○	○
Our first prompt v1 (Test #2)	●	◐	◐	○	○
Khanmigo, custom GPTs *	●	?	?	◐	○
Socratic AI (this project)	●	●	●	●	●

● yes ◐ partly ○ no ? not shown

** from public descriptions, not tested by us*

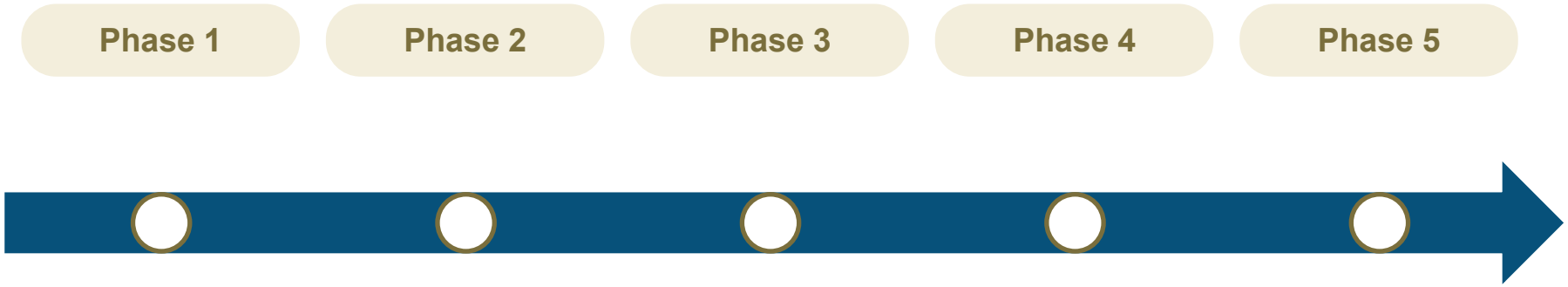
No tested tool has a full question ladder, framework marks and a clear session, and none shows the type of reasoning.

LITERATURE REVIEW

RESEARCH GAP

- AI can make students think less, and most AI tutors just give answers.
- Socratic AI tutors ask questions, but in no clear order.
- Most studies use what students say, not measured reasoning.
- No tool shows the type of reasoning or knowledge used.

PROPOSED METHODOLOGY



Early testing
Two Gemini Gem tutors tested; problems written down

Requirements
9 design requirements R1 to R9 from the tests

Build
AI agents, rules, cases, simulations, web app

Test and fix
Real sessions read turn by turn; problems fixed

Evaluate
Requirements, Assessor vs humans, learners

R1 Clear mission	R2 Fixed rounds, progress	R3 Remember, never re-ask
R4 Feedback and teaching	R5 Notice frustration	R6 Vary formats
R7 Separate assessor	R8 Learner stats	R9 Stable topic

Software
- Python 3.13, FastAPI, LangGraph
- Free Gemini 3.5 Flash Lite now, Claude AI supported
- HTML and JavaScript front end
- Supabase database, Render hosting (free)
- 196 automatic tests with pytest

Hardware
- A normal laptop, no GPU needed

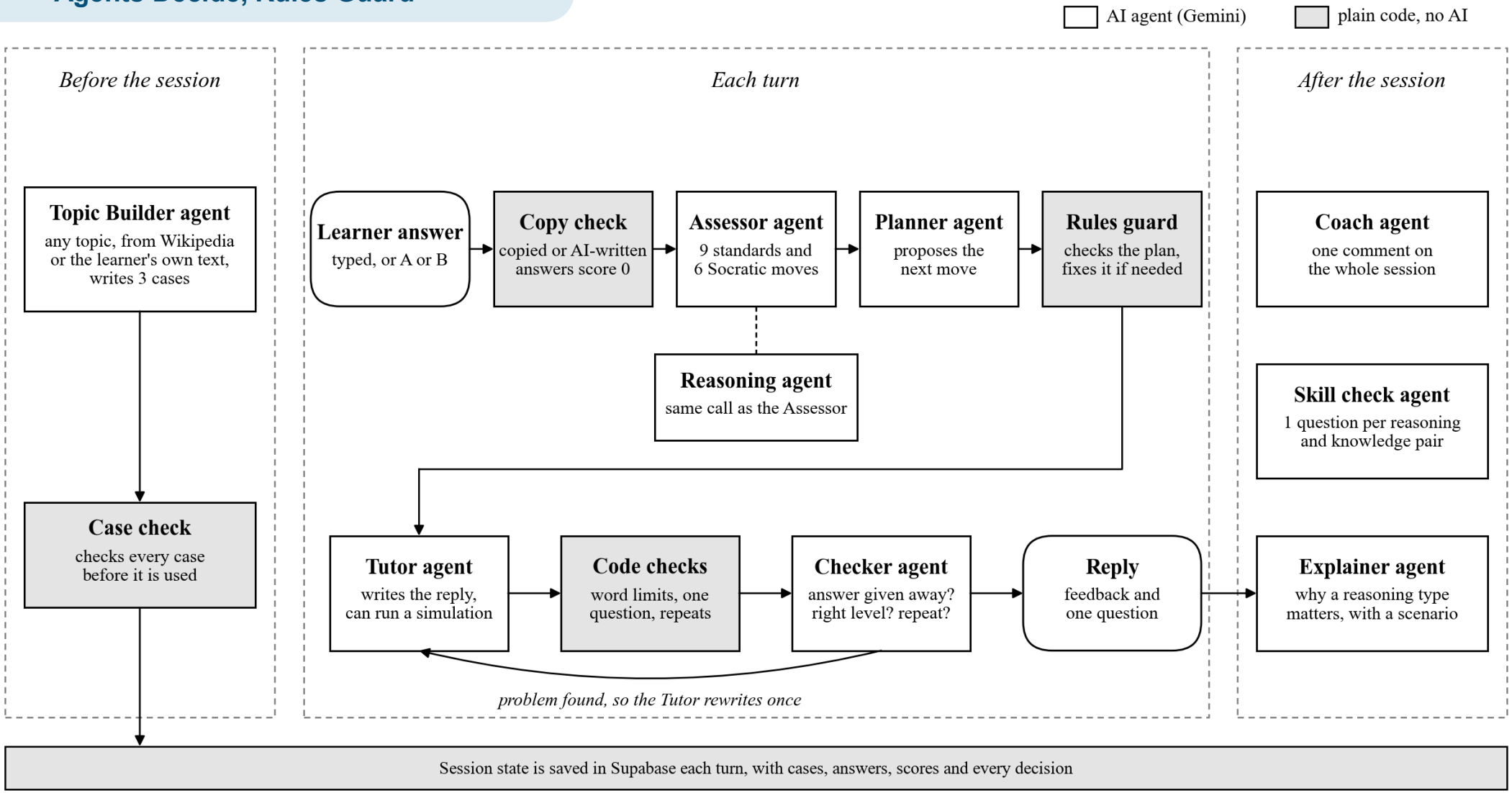
PROPOSED METHODOLOGY

WHY THIS DESIGN

- The Tutor follows Paul's six levels, so questions go up in a set order.
- A separate Assessor marks each answer on nine standards with evidence, so we can compare it with human markers.
- Code rules check every AI decision, so each step is fair and testable.
- A Reasoning agent now marks the type of reasoning and knowledge.

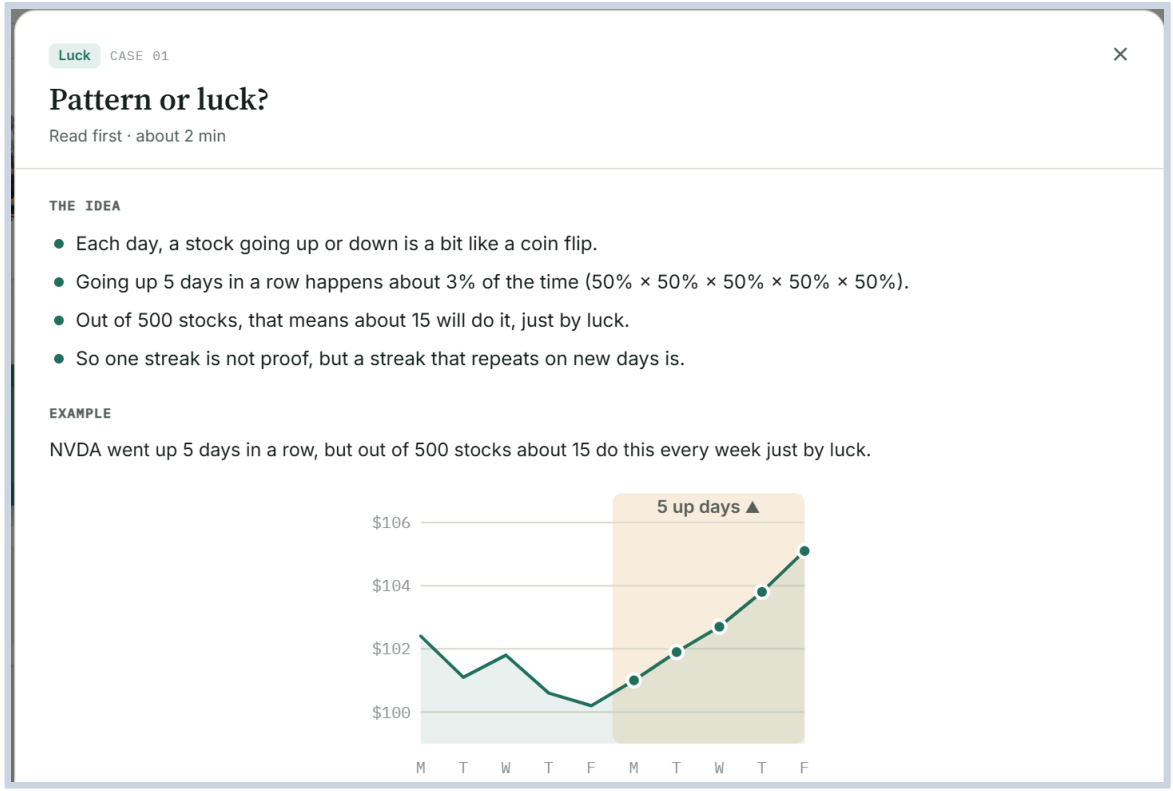
PROPOSED METHODOLOGY

Agents Decide, Rules Guard

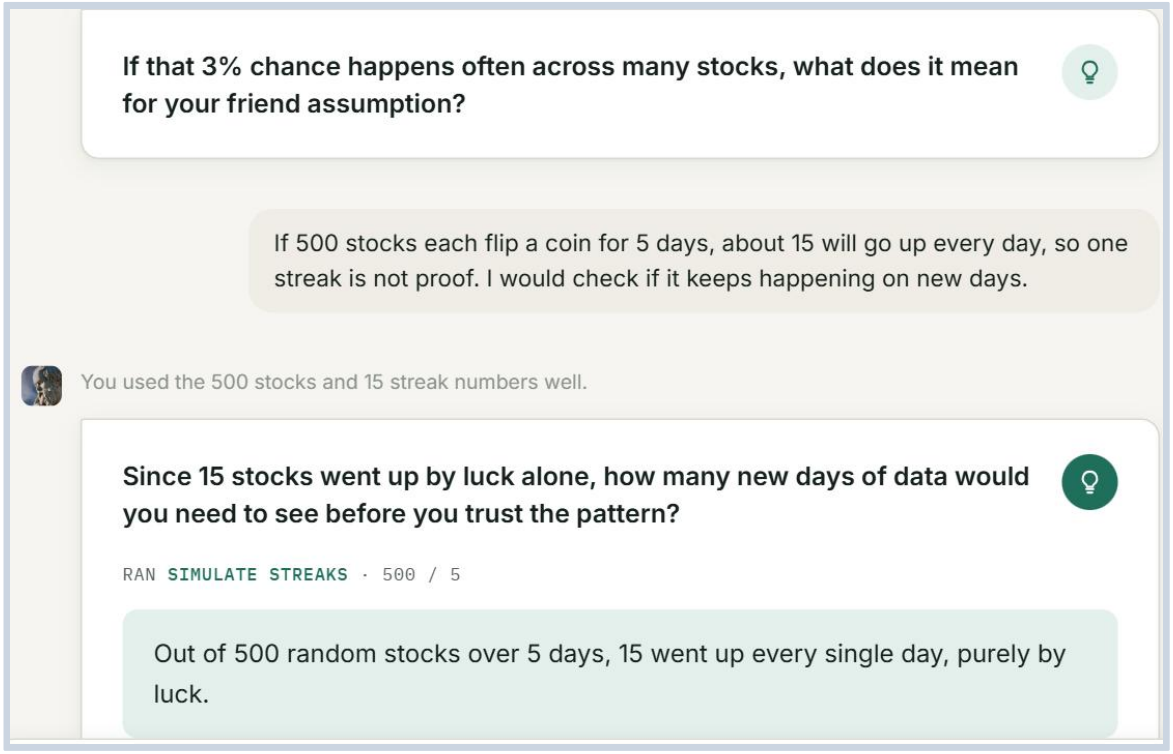


PROPOSED METHODOLOGY

Session Design (R1 to R9)



1. A short reading and a mission before each case

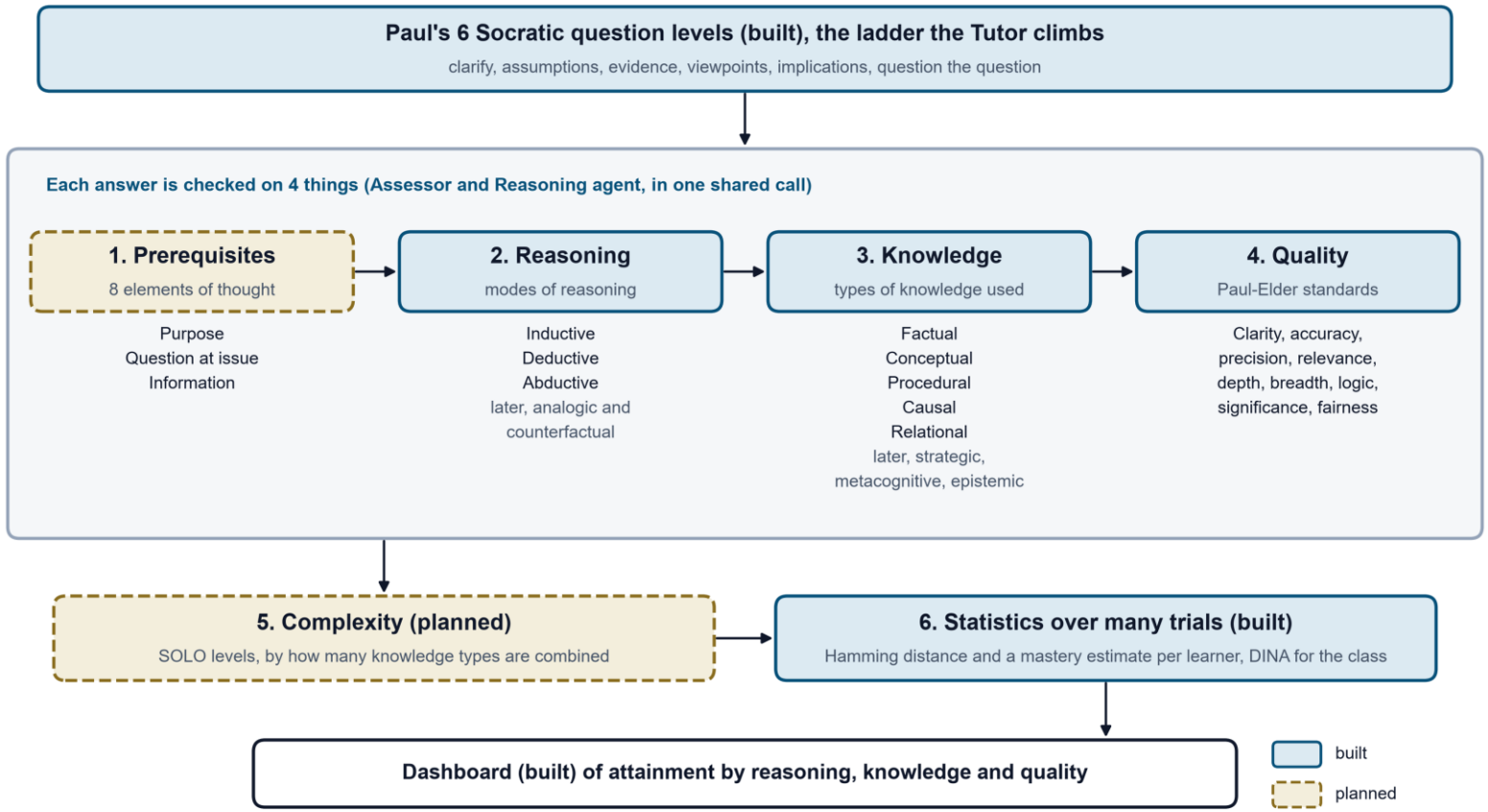


2. Feedback, one question, a hidden hint, and a simulation result

3, 6 or no-limit rounds | **Round and level shown** | **4 question formats** | **"I'm not sure" button** | **Status panel** | **Tutorial**

PROPOSED METHODOLOGY

Reasoning and Knowledge Layer (Goal 1, built)



- **The Tutor climbs Paul's 6 levels.** The student answers each question.
- **1 to 4 check each answer.** Basics, reasoning, knowledge and quality each get a 1 or 0.
- **Complexity (planned)**. One knowledge type is simple, and three joined is complex.
- **Statistics**. Many answers show which skills are mastered, and the Hamming distance [13] counts the skills left.

For example, "All the regional banks fell, not just CIMB" is inductive reasoning on relational knowledge.

PROPOSED METHODOLOGY

Evaluation Plan

1 Requirement check

Objective 1

Run the same cases on the Gem tutors and on Socratic AI, and mark R1 to R9.

2 Assessor vs human markers

Objective 2

Mark 61 answers by hand and compare them using Cohen's kappa (target 0.8).

3 With and without agents

All objectives

Switch the Planner, simulations or Checker off, and compare the same sessions.

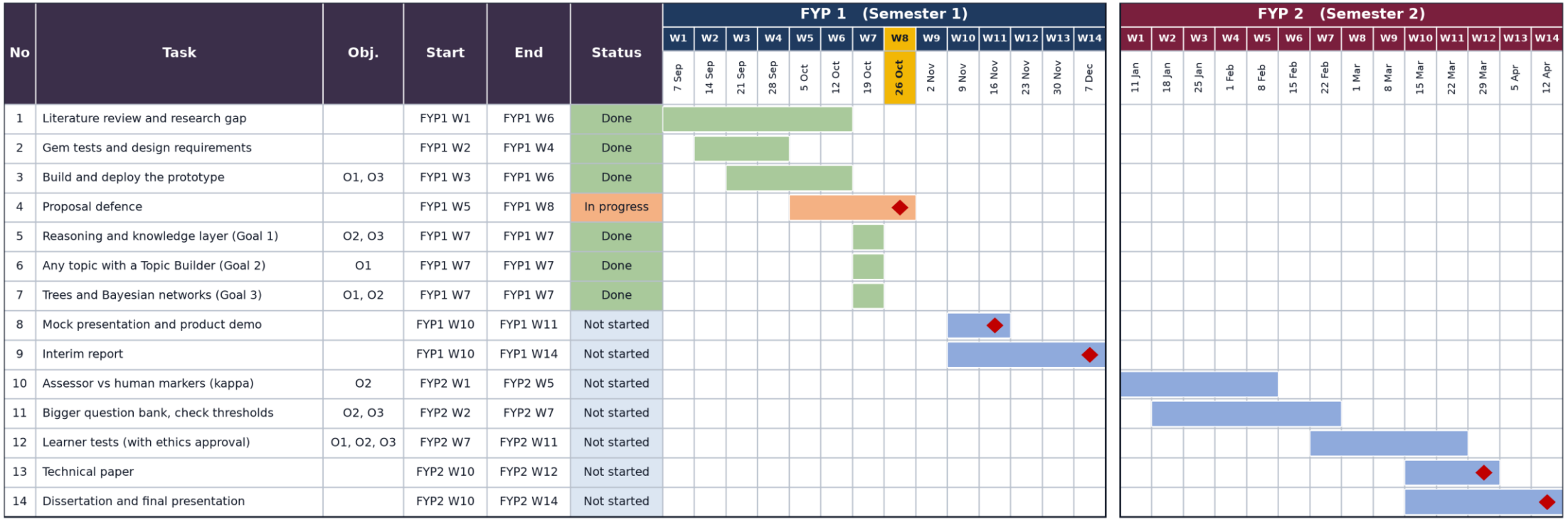
4 Learner experience

Objective 3

A short questionnaire with Dr Ho's students, after ethics approval.

GANTT CHART & MILESTONES

No	Task	Obj.	Start	End	Status
1	Literature review and research gap		FYP1 W1	FYP1 W6	Done
2	Gem tests and design requirements		FYP1 W2	FYP1 W4	Done
3	Build and deploy the prototype	O1, O3	FYP1 W3	FYP1 W6	Done
4	Proposal defence		FYP1 W5	FYP1 W8	In progress
5	Reasoning and knowledge layer (Goal 1)	O2, O3	FYP1 W7	FYP1 W7	Done
6	Any topic with a Topic Builder (Goal 2)	O1	FYP1 W7	FYP1 W7	Done
7	Trees and Bayesian networks (Goal 3)	O1, O2	FYP1 W7	FYP1 W7	Done
8	Mock presentation and product demo		FYP1 W10	FYP1 W11	Not started
9	Interim report		FYP1 W10	FYP1 W14	Not started
10	Assessor vs human markers (kappa)	O2	FYP2 W1	FYP2 W5	Not started
11	Bigger question bank, check thresholds	O2, O3	FYP2 W2	FYP2 W7	Not started
12	Learner tests (with ethics approval)	O1, O2, O3	FYP2 W7	FYP2 W11	Not started
13	Technical paper		FYP2 W10	FYP2 W12	Not started
14	Dissertation and final presentation		FYP2 W10	FYP2 W14	Not started



28 Oct 2026 (FYP1 W8)
Proposal defence

16 Nov 2026 (FYP1 W11)
Product demo, mock presentation

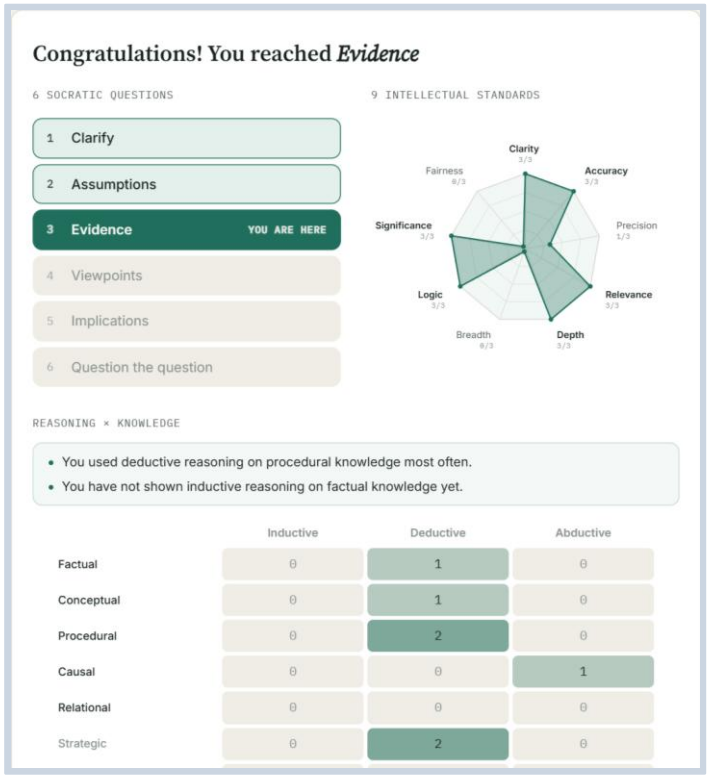
7 Dec 2026 (FYP1 W14)
Interim report

29 Mar 2027 (FYP2 W12)
Technical paper

12 Apr 2027 (FYP2 W14)
Dissertation, final presentation

PRELIMINARY RESULTS

A Real 3-Round Session (Gemini)



The end profile shows the level, the 9 standards and the reasoning and knowledge grid

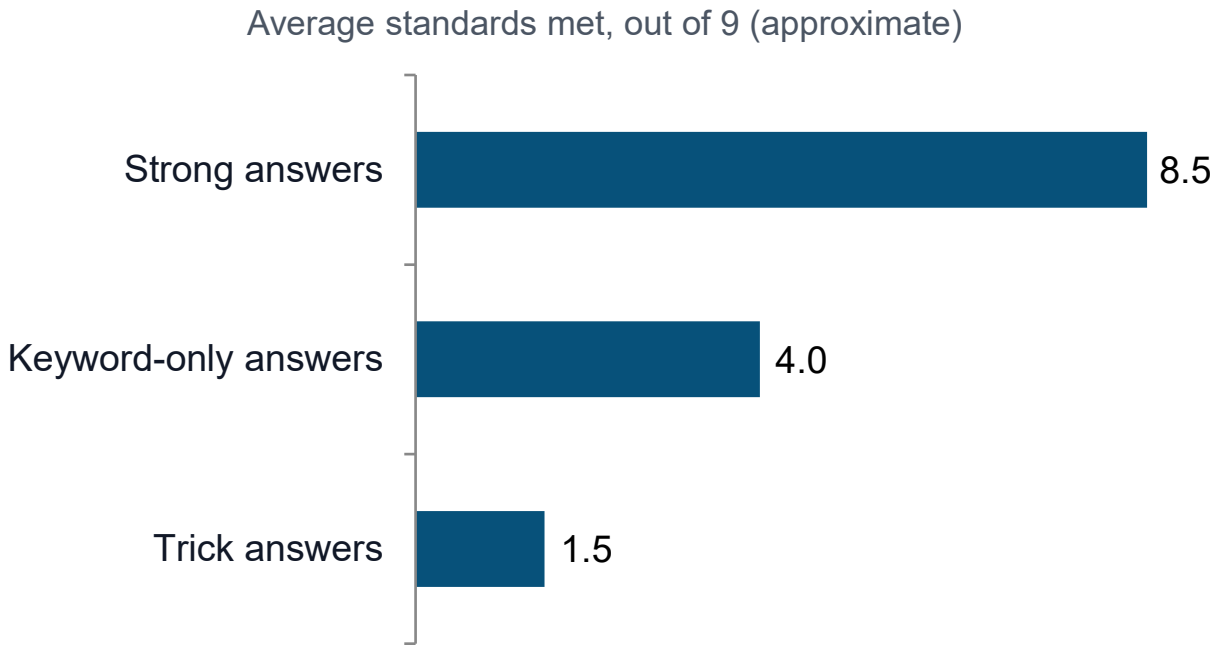
0 times the answer was given away, even when I asked for it

3 of 3 main problems found in testing were fixed

~24 AI calls per session, inside the free tier

PRELIMINARY RESULTS

Assessor on a 61-Answer Test Set



37 real and 24 made-up answers, not yet checked against human markers.

Main problems found and fixed

- The tutor asked the same question in new words, so the Checker now compares questions.
- Levels rose too fast, so a level up now needs 2 good answers.
- The Assessor was too generous, so each standard now has strict rules.

Next, keyword-only answers still score about 4 of 9, so I will make the Assessor stricter.

CONCLUSIONS

The Socratic AI tutors I tested have no **clear question order**, no **measurement** and no **adaptation**.

Socratic AI fixes all three. It asks up Paul's six levels, marks each answer on nine standards and adapts to each student.

It also shows the **type of reasoning** and **knowledge used** on any topic. Next, human markers will check its marks.

Take-home message

Socratic AI trains thinking, it is not an answer machine. AI agents work together, and simple code rules keep them in check.

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Q & A